```python
# Online Python compiler (interpreter) to run Python online.
# Write Pimport random as rand
import math
import time

lives = 100
points = 0

symbols1 = ("+","-")
symbols2 = ("*","/")
symbols3 = ("^", "√")

def space(x):
    for i in range(x):
        print("")

def randSymbol1():
    return symbols1[rand.randint(0,1)]

def randSymbol2():
    combinedSymbols1 = symbols1 + symbols2
    return combinedSymbols1[rand.randint(0,3)]

def randSymbol3():
    combinedSymbols2 = symbols1 + symbols2 + symbols3
    return combinedSymbols2[rand.randint(0,5)]
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def miniMax(mini,maxi):
    return mini, maxi

def mathProblems(grade):
    config = {
        1: {"symbol": lambda: "+",  "min": 1, "normMax":10},
        2: {"symbol": randSymbol1,  "min": 1, "normMax":20},
        3: {"symbol": randSymbol1,  "min": 1, "normMax":50},
        4: {"symbol": randSymbol2,  "min": 1, "normMax":100, "advMax":12},
        5: {"symbol": randSymbol3,  "min": 1, "normMax":150, "advMax":20, "powMax":5, "exp":2},
        6: {"symbol": randSymbol3,  "min": 1, "normMax":300, "advMax":35, "powMax":9, "exp":3},
        7: {"symbol": randSymbol3, "min": 2, "normMax": 500, "advMax": 50, "powMax":9, "exp":3},
        8: {"symbol": randSymbol3, "min": 5, "normMax":750, "advMax": 85, "powMax": 12, "exp":4}}

    return config[grade]


def genNumbers(symbol,grade): #number generator that ensures no decimals and no negatives

    mini = grade["min"]
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    if symbol in ("+", "-"):
        maxi = grade["normMax"]

    elif symbol in ("*", "/"):
        maxi = grade["advMax"]

    elif symbol in ("^", "√"):
        maxi = grade["powMax"]
        exp = grade["exp"]

    if symbol == "/":
        num2 = rand.randint(mini, max(2, maxi)) #doesnt make it divide by 0 since the limit is 2 to prevent undefined
        num1 = num2 * rand.randint(1, maxi // num2) #makes num1 a factor of num2 so no decimals

    elif symbol == "-":
        num1 = rand.randint(mini,maxi)
        num2 = rand.randint(mini,num1) #makes the num2 always smaller or equal to num1, preventing negatives

    elif symbol == "^":
        num1 = rand.randint(mini,maxi)
        num2 = exp
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    config = mathProblems(grade)
    symbol = config["symbol"]()
    num1, num2 = genNumbers(symbol, config)
    correct = calculate(num1, num2, symbol)

    print(f"{num1} {symbol} {num2} = ?\n")
    while attempts != 0:
        try:
            answer = int(input(">>     "))
        except ValueError:
            print("Uhm.. What? Input the right format of answer\n")
            continue

        if answer == correct:
            state['points'] += (1*multiplier)
            print("")
            print("Verdict: Correct!")
            print(f"Points: {state['points']}\n")
            break
        else:
            attempts -= 1
            state['points'] -= 1
            print("")
            print("Verdict: Wrong!")
            print(f"Attempts left: {attempts}\n")
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        print("")
        if attempts == 0:
            state['lives'] -= (10)
            print(f"Lives: {state['lives']}\n")

def runGame(playerName, state): #so that each person has their unique stuff

    print(f"Welcome, {playerName}! Let's begin.\n")

    choice = 0
    while state["lives"] > 0 and choice != 2:
        print("1. Play")
        print("2. Quit to menu")

        try:
            choice = int(input("Enter your choice.. >> "))

        except ValueError:
             print("I cant read this format..\n")
             continue

        if choice < 1 or choice > 2:
            print("Thats not in the choices..\n")
            continue
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    elif choice == 2:
        break


    #Stage 1 place holder
    space(2)
    if choice == 1:
        space(5)
        print("Once upon a time, in a dorm far far away, you were studying all\n")

        print("night and became thirsty while cramming for a math test tomorrow.\n")

        time.sleep(2)

        print("You decided to get up and grab a glass of fresh coffee.\n")


        print("As you made your way back to your desk...\n")
        time.sleep(2)
        print(",you tripped over a piece of scratch paper you\n")
        print("were using to vent out your mathematical stress ..\n")
        space(2)
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        time.sleep(2)
        print("and your head hit the table!, causing you to fall unconscious.\n")

        space(1)
        print("Once you woke up again, you were in a math dungeon,")
        print("where you had to answer math problems to escape.")


        space(2)

        print("Good luck..")

        space(3)

        while True:
            print("SELECT DIFFUCLTY\n")
            print("1. GRADE 1")
            print("2. GRADE 2")
            print("3. GRADE 3")
            print("4. GRADE 4")
            print("5. GRADE 5")
            print("6. GRADE 6")
            print("7. GRADE 7")
            print("8. GRADE 8\n")
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                try:
                    gradeSelect = int(input("Choose your fate: ))
                except ValueError:
                    print("What kind of grade is that..?\n")
                    continue

                if gradeSelect < 1 or gradeSelect > 8:
                    print("That's not in the range..\n")
                    continue

                if gradeSelect == 1:
                    print("Going easy huh? \n")
                    modifier = 1
                    break

                elif gradeSelect == 2:
                    print("Interesting\n")
                    modifier = 1
                    break

                elif gradeSelect == 3:
                    print("Thats okay\n")
                    modifier = 2
                    break

                elif gradeSelect == 4:
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                elif gradeSelect == 4:
                    print("Going harder?\n")
                    modifier = 3
                    break

                elif gradeSelect == 5:
                    print("Get ready..\n")
                    modifier = 4
                    break

                elif gradeSelect == 6:
                    print("Thats a big mountain to climb..\n")
                    modifier = 5
                    break

                elif gradeSelect == 7:
                    print("Oh well..\n")
                    modifier = 6
                    break

                elif gradeSelect == 8:
                    print("You chose this.\n")
                    modifier = 7
                    break


            print(f"SELECTED: GRADE {gradeSelect}")

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            for _ in range(5):
                question(grade=gradeSelect, multiplier=modifier, attempts=3, state=state)

                space(2)
                if state["lives"] <= 0:
                    break

            if state["lives"] <= 0:
                print(">>   YOU LOST   <<\n")
                return state
            else:
                return state

choice2 = 1
playerTable = {}
while lives > 0 and choice2 != 3:

    print("------The Equation Chronicles------\n")
    space(2)
    print("1. New Game")
    print("2. Load Game")
    print("3. Exit\n")
    try:
        choice2 = int(input(">>  "))
    except ValueError:
        print("I cannot read what you're saying..\n")
        space(1)
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        continue

    if choice2 == 1:
        name = input("Name:      ").strip()
        username = input("Username:    ").strip()
        password = input("Password:    ").strip()

        if username in playerTable:
            print("Username already taken! Try loading your game instead\n")
            continue

        playerTable[username] = {
            "name": name,
            "username": username,
            "password": password,
            "points": 0,
            "lives": 100
        }

        state = {
            "points": playerTable[username]["points"],
            "lives": playerTable[username]["lives"]
        }


        state = runGame(name, state)
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        if state:
            if state["lives"] <= 0:
                del playerTable[username]
                print(f"ACCOUNT TERMINATED")
            else:
                playerTable[username]["points"] = state["points"]
                playerTable[username]["lives"]  = state["lives"]


    elif choice2 == 2:
        if not playerTable:
            print("No accounts this yet. Start a new game
                first^^\n")
            continue

        print("Please enter your account:\n")
        userCheck = input("Username:    ")
        passCheck = input("Password:    ")

        user = playerTable.get(userCheck)

        if not user or user["password"] != passCheck:
            print("Invalid username or password.\n")
            continue

        state = {
            "points": user["points"],
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            "points": user["points"],
            "lives": user["lives"]
        }

        state = runGame(user["name"], state)

        if state:
            if state["lives"] <= 0:
                del playerTable[userCheck]
                print(f"ACCOUNT TERMINATED\n")
            else:
                user["points"] = state["points"]
                user["lives"]  = state["lives"]

    elif choice2 < 1 or choice2 > 3:
        print("Uhm, what range are you picking?")

print("See you next time.")
```

```
------The Equation Chronicles------



1. New Game
2. Load Game
3. Exit

>>  1
Name:     Zijan
Username:   zgcabigao
Password:   cabigaozg
Welcome, Zijan! Let's begin.

1. Play
2. Quit to menu
Enter your choice.. >> 1
```

Once upon a time, in a dorm far far away, you were studying all

night and became thirsty while cramming for a math test tomorrow.

You decided to get up and grab a glass of fresh coffee.

As you made your way back to your desk...

,you tripped over a piece of scratch paper you

were using to vent out your mathematical stress..

and your head hit the table!, causing you to fall unconscious.

Once you woke up again, you were in a math dungeon,
where you had to answer math problems to escape.

Good luck..

```
SELECT DIFFUCLTY

1. GRADE 1
2. GRADE 2
3. GRADE 3
4. GRADE 4
5. GRADE 5
6. GRADE 6
7. GRADE 7
8. GRADE 8

Choose your fate:    1
Going easy huh?

SELECTED: GRADE 1
5 + 9 = ?

>>     14

Verdict: Correct!
Points: 1



2 + 9 = ?

>>     11
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2 + 9 = ?

>>     11

Verdict: Correct!
Points: 2



4 + 8 = ?

>>     12

Verdict: Correct!
Points: 3



6 + 7 = ?

>>     13

Verdict: Correct!
Points: 4
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10 + 9 = ?

>>     19

Verdict: Correct!
Points: 5



------The Equation Chronicles------



1. New Game
2. Load Game
3. Exit

>>  2
Please enter your account:

Username:   zgcabigao
Password:   cabigaozg
Welcome, Zijan! Let's begin.

1. Play
2. Quit to menu
Enter your choice.. >> 2
```

```
1. New Game
2. Load Game
3. Exit

>>  3
See you next time.

=== Code Execution Successful ===
```